TEST REPORT

US IEC 60335-2-45:2002;US IEC 60335-1:2010
Household and similar electrical appliances – Safety

Report Number ..... :	**TR22061969-S-001**
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Date of issue ..... :	2026-6-19
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Testing laboratory name ..... :	**Guangdong Keyway Testing Technology Co., Ltd.**
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Applicant's name ..... :	NEWWAY TECHNOLOGY(SUZHOU) CO., LIMITED
Address ..... :	No. 20 Dagang Road, Fuqiao TownTaicang City, China
Manufacturer's name..... :	NEWWAY TECHNOLOGY(SUZHOU) CO., LIMITED
Address ..... :	No. 20 Dagang Road, Fuqiao TownTaicang City, China
Test specification:	
Standard ..... :	US IEC 60335-2-45:2002;US IEC 60335-1:2010
Procedure deviation ..... :	N/A
Non-standard test method ..... :	N/A
Test Report Form No. ..... :	IEC 60335-2-45;IEC 60335-1

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Test Item description ..... :	Plastic tube welding tools
Trade Mark ..... :	INGCO
Manufacturer ..... :	Same as applicant
Model/Type reference ..... :	PTWT215002
Batch No..... :	6928073604366
Ratings ..... :	220-240V, 50/60Hz, 800/1500W

Summary of construction and testing:

Whether parts of tests for the product have been subcontracted to other labs:

☐ Yes ☒ No

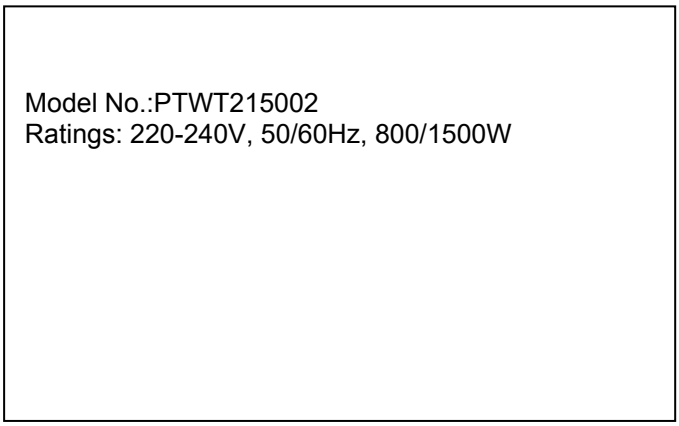
If Yes, list the related test items and lab information:

Test items:

Lab information:

Copy of marking plate:

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- This is a representative label, the other models are identical to it except for the model name.

POSSIBLE TEST CASE VERDICTS:	
- test case does not apply to the test object..... :	N/A
- test object does meet the requirement ..... :	P (Pass)
- test object does not meet the requirement ..... :	F (Fail)
TESTING:	
Date of receipt of test item..... :	2026-6-11
Date (s) of performance of tests..... :	2026-6-11 to 2026-6-19
GENERAL REMARKS:	
"(See Enclosure #)" refers to additional information appended to the report. **"(See appended table)" refers to a table appended to the report.** **Throughout this report a ☐ comma / ☒ point is used as the decimal separator.**	

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.	**General requirements**		**P**
4.1	The risks associated with the hazards relevant to the electrical equipment shall be assess as part of the overall requirements for risk assessment of the machine		P
4.2	Selection of equipment		P
	Electrical components and devices shall be suitable for their intended use and shall conform to relevant IEC standards where such exist		P
4.3	Electrical supply		P
	The electrical equipment shall be designed to operate correctly with the relevant conditions of supply	They can be operated correctly with the relevant conditions of supply	P
4.4	Physical environment and operating conditions		P
	Shall be suitable for use as specified:		P
	- electromagnetic compatibility	This machine is suitable for use as specified in this clause	P
	- ambient air temperature		
	- humidity		
	- altitude		
	- contaminants		
	- longizing and non-ionizing radiation		
	- vibration, shock and bump		
4.5	Transportation and storage		P
	-2.5℃ to + 55℃ and short periods not exceeding 24h at up to + 70℃		P
4.6	Provisions for handling		P
	Heavy and bulky equipment shall be moved by cranes or similar equipment		P
4.7	Instaliation and operation		P
	According to supplier’s instructions		P

5	**Incoming supply conductors terminations and devices for disconnecting and switching off**		P
5.1	Incoming supply conductor terminations		P
	Single or multiple power supply		P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The supply conductors are terminated at the supply disconnection device if not, the separate terminals shall be provided	Terminated at the supply disconnection device	P
	If a neutral conductor is used, it shall be indicated clearly in the technical documentation	Not applicable Not neutral has been used	N
	Labelled N shall be provided for the neutral conductor	Neutral has been used	P
	No connection between the protective bonding circuit and the neutral conductor		P
	All terminals for the incoming suply connection shall be identified clearly	All of them have been identified clearly	P
5.2	Terminal for connection to the external protective earthing system		P
	Shall be in the vicinity of the associated phase conductor terminals		P
	Cross-sectional area of the external protective copper conductor according to table 1		N
	Marking of the external protective conductor with the letters ”PE”		N
	Marking of earthing terminals		P
5.3	Supply disconnecting (isolating) device		P
5.3.1	A supply disconnecting device shall be provided:		P
	- for each incoming source of supply to a machine		P
	- for the source of supply to a feeder system using collector wires, collector bars, slip-ring assemblies, flexible cable systems, to a machine or a number of machines		P
	- for each on-board power supply		N
	When two or more supply disconnecting devices are provided, protective interlocks for their correct operation shall be used where a hazardous condition or damage to the machine or to the work in progress can occur	Not applicable	N
5.3.2	The supply disconnecting device shall be one of the following types:		P
	a) a switch-disconnector, with or without fuses, in accordance with IEC 60947-3, utilization category AC-23B or dc-23B	Pass muster	P
	b) a switch-disconnector, with or without fuses, in accordance with IEC 60947-3, that has an auxiliary contact that in all cases causes switching devices to break the load circuit before the opening of the main contacts of the disconnector		N

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	c)a circuit-breaker suitable for isolation in accordance with IEC 60947-2		P
	d)a plug/socket combination for a machine with a rated current not exceeding 16A and a total power rating not exceeding 3KW		N
	e) a plug/socket-outlet or an appliance coupler for a flexible cable supply to a mobile machine under the following condition:		N
	- it shall not be possible to connect or disconnect a ply and socket-outlet or an appliance couplet, without breaking capacity, during load conditions		N
	- the plug and socket-outlet or the appliance coupler shall be so connected that the part connected to the incoming supply is that which is protected to at least IP2X or IPXXB		N
	The electrical equipment shall have a device for switching the machine on and off		P
5.3.3	Requirements		P
	Have one OFF and one ON position only		P
	Marked clearly with ”I” and ”O”		P
	Have a reset (tripped) position between ”O” and ”I”		P
	Have an external operating handle		N
	The handle should be Black or Grey		N
	Could be locked in the OFF position		P
	Disconnect all live conductors of its power supply circuit		P
	Sufficient breaking capacity		P
5.3.4	Operating handle		N
	Shall be easily accessible and located:0.6m~0.9m	Not such handle	N
5.3.5	Excepted circuits		N
	Have their own disconnecting device (Reconnended)	Not applicable No excepted circuit has been found	N
	If no disconnecting device, the relevant safety requirements shall be complied with	Not applicable No excepted circuit has been found	N
5.4	Devices for switching off for prevention of unexpected start-up		P
	Unexpected start-up shall be prevented (Devices described in 5.3.2 may fulfil this function)	Pass muster	P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
5.5	Devices shall be provided for disconnecting (isolating) electrical equipment to enable work to be carried out without a risk form electric shock or burn		P
5.6	Protection against unauthorized, inadvertent and/or mistaken connection		P
	The devices described in 5.4 and 5.5 shall be equipped with such function		P
6	**Protection against electric shock**		P
6.1	General	See the relevant clauses	P
6.2	Protection against direct contact		P
6.2.1	General		P
	Either 6.2.2 or 6.2.3 and, where applicable, 6.2.4 shall be applied	See the relevant clauses	P
	When the equipment is located in places open to all persons, measures of either 6.2.3 or 6.2.2 with a min, degree of protection against direct contact corresponding to IP4X or IPXXD shall be applied	Not applicable This machine shall be located in the factory, and be operated by the authorized persons	N
6.2.2	Protection by enclosures		P
	Min protection degree for live parts: IP2X or IPXXB		P
	Min protection degree for top surface: IP4X or IPXXD		P
	Opening an enclosure shall only be possible under one of the folowing conditions:		P
	a) the use of a key or tool is necessary bu skilled or instructed persons		P
	b) the disconnection of live parts inside the enclosure before the enclosure may be opened(Use of the supply disconnecting device)		P
	c) opening without the use of a key or a tool and without disconnection of live parts shall be possible only when the min.protection degree is IP2X or IPXXB	Not applicable No this kind of situation	N
6.2.3	Rotection by insulation of live parts		P
	Live parts shall be covered by insulation which can only be removed by destruction		P
	Such insulation shall withstand the mechanical, chemical, electrical and thermal stresses under normal service conditions	Pass muster	P
6.2.4	Protection against residual voltages		N

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	After disconnecting, any exposed conductive part having a residual voltage that shall be discharged to 60V or less within 5 seconds	Not exposed conductive part	N
	If mentioned above is not possible, a warning notice drawing shall provided	Not applicable	N
	If the withdrawal of plugs or similar devices would make the exposure of the conductors, the discharge time shall not exceed I second such conductor shall have the protection degree at least IP2X or IPXXB	Not applicable No this kind of situation has been found	N
6.2.5	Protection by barriers	No barriers	N
	For protection by barriers, see 412.4 or IEC60364-4-41		N
6.2.6	Protection by placing out of reach or protection by obstacles		N
	For protection by placing out of reach see 412.4 of IEC60364-4-41		N
	For protection by obstacles see 412.4 of IEC60364-4-41		N
	For collector wire systems or collector bar systems with a degree of protection less than IP2X see 13.8.1		N
6.3	Protection against indirect contact		P
6.3.1	General		P
	For each circuit or part, at least one of the measures in accordance with 6.3.2 to 6.3.3 shall be applied	See the following descriptions	P
6.3.2	Measure to prevent the occurrence of a hazardous touch voltage		P
6.3.2.1	General		P
6.3.2.2	Protection by use of class II equipment or by equivalent insulation		N
	Application of class II equipment or equivalent insulation		N
6.3.2.3	Protection by electrical separation		P
	Application of electrical separation		P
6.3.2.4	Supply system design		N
	Application of a supply system designed with its neutral point either insulated from or having a high impedance to earth	Not applicable	N

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
6.3.3	Protection by automatic disconnection of supply		N
	Use of the automatic disconnection of supply		N
6.4	Protection by the use of PELV	No PELV circuit has been used	N
6.4.1	General requirements		N
	PELV (protective extra-low voltage) circuits shall satisfy all of the conditions specified in this clause		N
6.4.2	Sources for PELV		N
	The sources for PELV shall be one of the conditions specified in this clause		N
	Protection of equipment		N
7	**Protection of equipment**		**P**
7.1	General: the measures to be taken to protect equipment against the effects of :		P
	- overcurrent arising from a short circuit		P
	- overload current		P
	- earth fault		P
	- overvoltage due to lightning and switching surges		N
	- abnormal temperature		N
	- loss of or reduction in the supply voltage		P
	- overspeed of machines/machine elements		P
	- incorrect phase sequence		P
7.2	Over current protection		P
7.2.1	General		P
7.2.2	Supply conductors		P
	The supplier is not responsible for providing the over current device for the supply conductors		P
	Installation diagram with data necessary for selection of the over current protective device	Pass muster	P
7.2.3	Power circuits		P
	All conductors shall be protected against over current (except earthed neutral conductor)		P
	Cross-section area of neutral conductor	No neutral conductor has been used	N
	For neutral conductors smaller than phase conductors then IEC 364-4-473 shall apply	No neutral conductor has been used	N
	In IT-systems, it is recommended that the neutral conductor is not used	Not applicable	N
7.2.4	Control circuits		P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Conductors of control circuits connected to the supply voltage and of circuits feeding control circuit transformers shall be protected against over current in accordance with 7.2.3	Pass muster	P
7.2.5	Socket outlets and their associated conductors		P
	Over current protection devices shall be provided in the unearthed live conductors		P
7.2.6	Lighting circuits		N
	All unearthed conductors of circuits supplying lighting shall be protected against the effects of short circuits by the procision of over current devices separate from those protecting other circuits		N
7.2.7	Transformers		P
	Transformers shall be protected aainst Over current in accordance with IEC60076-5 and IEC60743 as appropriate	All transformer have been protected against over current	P
	The type and setting of the overcurrent protective device should be in accordance with the recommendations of the transformer supplier		P
7.2.8	Location of over current protective device		P
	Over current protective device shall be located at the point where the conductors to be protected are connected to their supply	Pass muster	P
7.2.9	Over current protective devices		P
	Sufficient breaking capacity		P
	Where fuses are used, a type readily available in the country of use shall be selected, or arrangement shall be made with the use for the supply of spare parts	Pass muster	P
7.2.10	Rating and setting of over current protective devices		P
	The rated current of fuses or the setting current of other over current protective devices shall be selected as low as possible but adequate for the anticipated over currents		P
	The rated current or setting of an over current protective device is determined by the current carrying capacity of the conductors to b e protected by that device in accordance with 13.4		P
7.3	Over load protection of motors		P
	Overload protection of motors shall be provided for each motor rated at more than 0.5KW	The overload protection is provided	P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict

	In applications where an automatic interruption of the motor operation is unacceptable, the overload detection shall give a warning signal to which the operator can respond	Pass muster	P
	Detection of overload shall be provided in each live conductor excepted for the neutral conductor	This function has been provided	P
	For motors having single-phase or d.c. power supplies. Detection in only one unearthed live conductor is permitted		P
	Automatic restarting of any motor after the operation of overload protection shall be prevented	No automatic restarting function has been used	P
7.4	Abnormal temperature protection		N
	Use of abnormal temperature protection	No used	N
7.5	Protection against supply interruption or voltage reduction and subsequent restoration		P
	Where a voltage drop or a supply interruption can cause a hazardous condition, damage to the machine, or to the work in progress, under voltage protection shall be provided		N
	The operation of the under voltage device shall not impair the operation of any stopping control of the machine		N
	Upon restoration of the voltage or upon swiching on the incoming supply, automatic or unexpected restarting of the machine shall be prevented	Automatic of unexpected restarting of the machine can be prevented	P
	Where only a part of the machine or of the group of machines working together in a coordinated manner is affected by the voltage reduction or supply interruption, the under voltage protection shall initiate appropriate control responses to ensure co-ordination+		N
7.6	Motor over speed protection		P
	Use of the motor over speed protection		P
7.7	Earth fault/residual current protection		P
	Use of earth fault/residual current protection for automatic disconnection		P
7.8	Phase sequence protection		P
	Where an incorrect sequence of the supply voltage can cause a hazardous condition or damage to the machine protection shall be provided		P
7.9	Protection against over voltage due to lighting and to switching surges		N
	Protection devices can be provided to protect against the effects of over voltages due to lighting or to switching surges		N

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Devices for the suppression of overvoltages due to lightning shall be connected to the incoming terminals of the supply disconnecting device		N
	Devices for the suppression of overvoltages due to switching surges shall be connected across the terminals of all equipment requiring such protection		N
8	**Equipotential bonding**		P
8.1	This clause provides requirements for both protective bonding and operational bonding		P
8.2	Protective bonding circuit		P
8.2.1	General		P
	On mobile machines with on-board power supplies, it shall be connected to a protective bonding terminal to provide protection against electric shock	Pass muster	P
	When a mobile machine is laso cppable of being connected to an external incoming supply, the protective bonding terminal shall be the connection point for the external protective conductor	Pass muster	P
	All parts of the protective bonding circuit shall be so designed that they are capable of with standing the highest thermal and mechanical stresses	This requirement has been complied with	P
	Any structural part of the electrical equipment or of the machine may be used as part of protective bonding circuit	They have been used as part of protective bonding circuit	P
	If an IT distribution system is used, the machine structure shall be sued as part of the protective bonding circuit in conjunction with an earth fault supervision system	Not applicable	N
	The structural bonding is not required where all the equipment provided is in accordance with 6.3.2.2		P
8.2.2	Protective conductors		P
	Protective conductiors shall be identified according to 14.2.2		P
	Copper conductors should be used		P
	Where a conductors material other than copper is used, its electrical resistance per unit length shall not exceed that of the allowable copper conductor and such conductors shall not be less than 16mm^2 in cross-sectional area	Pass muster	N

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The cross-sectional area of protective conductors shall be determined according to the requirements of : - 543 of IEC60364-5-54 or - 7.4.3.1.7 of IEC 60439-1, as appropriate	They have been used according to these requirements	P
8.2.3	Continuity of the protective bonding circuit		P
	All exposed conductive parts of the electrical equipment and machine shall be connected to the protective bonding circuit	connected to the protective bonding circuit	P
	Connection and bonding points shall be so designed that their current-carrying capacity is not impaired by mechanical ,chemical, or electrochenical influences	Copper	P
	Metal ducts of flexible or rigid construction and metallic cable sheaths shall not be used as protective conductors	No use metal ducts of flexible or rigid construction and metallic cable sheaths	P
	The continuity of the protective conductor in cables that are exposed to damage shall be ensured by appropriate measures	No such equipment	N
8.2.4	Exclusion of switching devices from the protective bonding circuit		P
	The protective bonding circuit shall not incorporate a switch device, an overcurrent protective device nor a means for current detection for such devices		P
8.2.5	Parts that need not be connected to the protective bonding circuit		P
	This applies to small parts such as screws, rivets, and nameplates and to parts inside an enclosure, irrespective of their size		P
8.2.6	Interruption of the protection bonding circuit		P
	Where the continuity of the protective bonding circuit can be interrupted by means of removable current collectors or plug/socket combinations, the protective bonding circuit shall not be interrupted before the live conductors have been disconnected, and shall be re-established before any live conductor is reconnected		P
	Metallic housings of plug/socket combinations shall be connected to the protective bonding circuit except where used for PELV	No such equipment	N
8.2.7	Protective conductor connecting points		P
	All protective conductors shall be terminated in accordance with 14.1.1		P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Each protective conductor connecting point shall be identified as such using the symbol 60417-2-IEC-5019" ⏚"		P
	Terminals for connection of the protective conductor may be identified by the bicolour combination GREEN-AND-YELLOW		P
8.3	Bonding for operational purposes		P
8.3.1	The objective of operational bonding is to minimize:		P
	-the consequence of an insulation failure on the operation of the machine		P
	-the consequences of electrical disturbances on the operation of sensitive electrical equipment		P
8.3.2	Bonding to the protective circuit		P
	One method for protection against unintended operation as a result of insulation failures is achieved by connecting one side of a control circuit fed by a transformer to the protective bonding circuit, with the control devices connected in accordance with 9.1.4	The measure described in this clause has been used	P
8.3.3	Bonding to common reference potential		P
	The effects of disturbances can be reduced by employing a low resistance conductor in a low impedance network that is used as a reference level for high frequency signals within the electrical equipment	Pass muster	P
9	**Control circuits and control functions**		P
9.1	Control circuits		P
9.1.1	Control circuit supply		P
	Transformers shall be used for supplying the control circuits	The transformers have been used	P
	Transformers are not mandatory for machines with a single motor starter and a maximum of two control devices	Not applicable	N
9.1.2	Control circuit voltages		P
	The nominal voltage shall not exceed 277V when supplied from a transformer		P
9.1.3	Protection		P
	Over current protection shall be provided according to 7.2.4 and 7.2.10	The over current protection has been provided	P
9.1.4	Connection of control devices		P
	Appropriate connection for control devices		P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
9.2	Control functions		P
9.2.1	Start functions		P
	Start functions shall operate by energizing the relevant circuit		P
9.2.2	Stop functions		P
	Each machine shall be equipped with appropriate stop functions		P
9.2.3	Operating modes		P
	Each machine can have one or more operating modes determined by the type of machine and its application	Pass muster	P
	Safeguarding shall remain effective for all operating modes		P
	Indication of the selected operating mode shall be provided		P
9.2.4	Suspension of safeguarding		P
	Where it is necessary to suspend safeguarding, a mode selection device or means capable of being secured in the desired mode shall be provided so as to prevent automatic operation	Pass muster	P
9.2.5	Operation		P
9.2.5.1	The necessary interlocks shall be provided for safe operation		P
	Measures shall be taken to prevent movement of the machine in an unintended manner after any stopping of the machine	No unintended operation can be occurred after any stopping of the machine	P
9.2.5.2	Start		P
	The start of an operation shall be possible only when all of the safeguards are in place and are functional except for conditions as described in 9.2.4		P
	On those machines where safeguards cannot be applied for certain operations, manual control of such operations shall be by hold-to-run controls, together with enabling devices, as appropriate	Not applicable	N
	Suitable interlocks shall be provided to secure correct sequential starting		P
	One machines requiring the use of more than one control station to initiate a start:		P
	- each control station shall have a separate manually actuated start control device		P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	- all required conditions for machine operation shall be met		P
	- all start control devices shall be in the released position before a start may be permitted		P
	- all start control devices shall be actuated concurrently		P
9.2.5.3	Stop		P
	Stop functions shall override related start functions		P
	The STOP function required for sewing units and systems is met by a hold-to-run control device		P
9.2.5.4	Emergency operations		P
9.2.5.4.1	General		P
9.2.5.4.2	Emergency stop		P
	Shall function either as a category o stop or as a category 1 stop	Category 1 stop	P
	The choice of the emergency stop shall be determined by the risk assessment of the machine	According to the result of risk assessment of the machine	P
	Where a category 0 stop is used for emergency stop function, it shall have only hard-wired electromechanical components		P
	The operation of emergency stop shall not depend on electronic logic or on the transmission of commands over a communications network or link	No this kind of situation	P
	Where a category 1 stop is used for the emrgency stop function, final removal of power to the machine actuators shall be ensured and carried out by means of electromechanical components	The final removal of power to the machine actuators is ensured by the controller and carried out by means of electromechanical components	P
9.2.5.4.3	Emergency switching off		P
	The functional aspects of emergency switching off are given in IEC60364-4-46		P
	Emergency switching off is accomplished by disconnecting the incoming supply of the machine effecting a category 0 stop	1 stop	N
	When a machine cannot tolerate the category 0 stop, it may be necessary to provide other protection ,for example against direct contact, so that emergency switching off is not necessary	1 stop	P
9.2.5.5	Monitoring of command actions		P
	Movement or action of a machine or part of a machine that can result in a hazardous condition shall be monitored	They have been monitored by the controller during the working of the machine	P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Movement or action of a machine or part of a machine that can result in a hazardous condition shall be monitored	They have been monitored by the controller during the working of the machine	P
9.2.5.6	Hold-to-run controls		P
	Hold-to-run controls shall require continuous actuation of the control device to achieve operation	Pass muster	P
9.2.5.7	Two-hand control		P
	Three types of two-hand control are available, the selection of which is determined by the assessment	Pass muster	P
9.2.5.8	Enabling device		P
	It shall be designed to allow motion when actuated in one position only (In any other position motion shall be stopped)	These machines have been designed to allow motion when actuated in one position only.	P
9.2.6	Combined start and stop controls		N
	Push-buttons and similar control devices that ,when operated, alternately initiate and stop motion shall only be used for functions which cannot result in a hazardous condition	Not applicable	N
9.2.7	Cableless control	No cableless control is used	N
9.2.7.1	General		N
	This subclause deals with the functional requirements of control systems employing cableless techniques for transmitting commands and signals between a machine control system and operator control station		N
	Means shall be provided to readily remove or disconnect the power supply of the operator control station		N
	Each operator control station shall carry an unambiguous to be controlled by that operator control station		N
9.2.7.2	Control limitation		N
	Measures shall be taken to ensure that control commands:		N
	- affect only the intended machine		N
	- affect only the intended functions		N
	Measures shall be taken to prevent the machine from responding to signals other than those from the intended operator control station		N
9.2.7.3	Stop		N
	Operator control stations shall include a separate and clearly identifiable means to initiate the stop function of the machine or of all the motions that can cause a hazardous condition		N

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
9.2.7.4	Serial data communication		N
	In a machine where the control of safety-related functions relies on serial data transfer, correct communications shall be ensured by using an error detection method that is able to cope with up to three error bits in any command sequence		N
9.2.7.5	Use of more than one operator control station		N
	Where a machine has more than one operator control station, measures shall be taken to ensure that only one control station can be enabled at a given time		N
9.2.7.6	Battery-powered operator control stations		N
	A variation in the battery voltage shall not cause a hazardous condition		N
9.3	Protective interlocks		P
9.3.1	Reclosing or resetting of an interlocking safeguard		P
	The reclosing or resetting of an interlocking safeguard shall not initiate machine motion or operation where that can result in a hazardous condition	No safeguard can initiate machine motion or operation	P
9.3.2	Overtravel limits		P
	Where an overtravel can cause a hazardous condition, a position sensor or limit switch shall be provided to initiate appropriate devices	Appropriate position sensors and limit switches have been used	P
9.3.3	Operation of auxiliary functions	Not applicable	N
	The correct operation of auxiliary functions shall be checked by appropriate devices		N
9.3.4	Interlocks between different operations and foe contrary motions		P
	All contactors, relays, and other control devices that control elements of the machine and that can cause a hazardous condition when actuated at the same time, shall be interlocked against incorrect operation	Pass muster	P
	Reversing contactors shall be interlocked in such a way that in normal service no short circuit can occur when switching		P
9.3.5	Reverse current braking		N
	Where reverse current braking is used on a motor, effective measures shall be taken to avoid the motor starting in the opposite direction at the end of braking where that reversal can cause a hazardous condition or damage to the machine or to the work in progress	Not applicable	N
9.4	Control functions in the event of failure		P
9.4.1	General requirements		P
	Provision of control functions in case of failure according to the level of risk assessment	According to the risk assessment	P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
9.4.2	Measures to minimize risk in the event of failure		P
9.4.2.1	Use of proven circuit techniques and components	Appropriate components have been used	P
9.4.2.2	Provisions for redundancy		N
	By providing partial or complete redundancy it is possible to minimize the probability that one single failure in the electrical circuit can result in a hazardous condition	Not applicable	N
9.4.2.3	Use of diversity		N
	The use of control circuits having different principles of operation or using different types of devise may reduce the probability of hazardous resulting from faults and/or failures	Not applicable	N
9.4.2.4	Functional tests		P
	Carried out automatically by the control system or manually by inspection	By inspection manually	P
9.4.3	Protection against maloperation due to earth faults, voltage interruptions and loss of circuit continuity		P
9.4.3.1	Earth faults		P
	Earth faults on any control circuits shall not cause unintentional starting, potentially hazardous motions, or prevent stopping of the machine	Make reference to the relevant clauses	P
9.4.3.2	Voltage interruptions		P
	Where a memory device is used, proper functioning in the event of power failure shall be ensured to prevent any loss of memeory taht can result in a hazardous condition	Any loss of memory can’t result in a hazardous condition	P
9.4.3.3	Loss of circuit continuity		N
	Where the loss of continuity of safety-related control circuits depending upon sliding contacts can result in a hazardous condition, appropriate measures shall be taken	Not applicable	N
10	**Operator interface and machine-mounted control devices**		**P**
10.1	General		P
10.1.1	General device requirements		P
	The clause contains requirements for devices mounted outside or partially outside control enclosures	These requirements appropriate for this machine have been complied with	P
10.1.2	Location and mounting		P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	As far as is practicable ,machine-mounted control devices shall be :		P
	- readily accessible for service and maintenance		P
	- mounted in such a manner as to minimize the possibility of damage from activities such as material handling		P
	The actuators of hand-operated control devices shall be selected and installed so that:		P
	- they are not less than 0,6m above the servicing level and are within easy reach of the normal working position of the operator		P
	- the operator is not placed in a hazardous situation when operating them		P
	- the possibility of inadvertent operation is minimized		P
10.1.3	Protection		P
	Where mounted as intended, operator interface and machine mounted control devices shall withstand the stresses use	withstand the stresses use	P
10.1.4	Position sensors		P
	Position sensors shall be so arranged that they will not be damaged in the event of overtravel	Because of the location of those position sensors, they can not be damaged in the event of over travel.	P
	Position sensors used in circuits with safety-related functions either shall have positive opening operation or shall provide similar reliability	Positive opening mode	P
10.1.5	Portable and pendant control stations		N
	Portable and pendant control stations and their control devices shall be so selected and arranged as to minimize the possibility of inadvertent machine operations caused by shocks and vibrations	Not applicable	N
10.2	Push-buttons		P
10.2.1	Colors		P
	Push-button actuators shall be color-coded according to table 2		P
	The colours for START/ON actuators should be WHITE,GREY or BLACK with a preference for WHITE. GREEN is also permitted. RED shall not be used	Pass muster	P
	The colour RED shall be used for emergency stop and emergency switching off actuators	Pass muster	P
	As far as is practicable, push-button actuators shall be colour-coded in accordance with table 2; limitations of the practicability are sizes of actuators and built-in casing, design of actuators		P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
10.2.2	Markings		P-
10.2.3	Use of adequate markings for push-buttons	Adequate markings are used	P
10.3	Indicator lights and displays		P
10.3.1	Modes of use		P
	Indication and / or confirmation		P
10.3.2	Colors		P
	Unless otherwise agreed between the supplier and the user, indicator light lenses shall be colour-coded with respect to the condition of the machine in accordance with Table 3	Their colors are according to table 3	P
10.3.3	Flashing lights		P
	Use of flashing lights		P
10.4	Illuminated push-buttons		N
	As far as is practicable, illuminated push-buttons shall be colour-coded in accordance with table 2 and table 3; limitations of the practicability are sizes of actuators and built-in casing, design of actuators	No such equipment	N
10.5	Rotary control devices	No such equipment	N
	Devices having a rotational member, such as potentiometers and selector switches, shall be mounted in such a way as to prevent rotation of the stationary member		N
10.6	Start devices		P
	Actuators used to initiate a start function or the movement of machine elements shall be constructed and mounted so as to minimize inadvertent operation	Flat type start push-buttons are used to prevent inadvertent operation	P
10.7	Devices for emergency stop		P
10.7.1	Location		P
	Devices for emergency stop shall be readily accessible	It is readily accessible	P
	Emergency stop devices shall be located at each operator control station and at other locations where the initiation of an emergency stop can be required	All of them are located at each operator control station	P
10.7.2	Type		P
	The type shall be of the self-latching type and shall have positive opening operation	A push-button operated switch	P
	Shall be of the self-latching type and shall have positive opening operation	Self-latching type and positive opening operation	P
10.7.3	Restoration of normal function after emergency stop		P
	It shall not be possible to restore an emergency stop circuit until the emergency stop device has been manually reset	This requirement has been complied with	P
10.7.4	Actuators		P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Actuators of emergency stop devices shall be coloured RED.	RED	P
	The background immediately around the actuator shall be coloured YELLOW		P
10.7.5	Local operation of the supply disconnecting device to effect emergency stop		P
	The supply disconnecting device may be locally operated to serve the function of emergency stop when:		P
	- it is readily accessible toe the operator		P
	- it is of the type described in 5.3.2 a) ,b) or c)		P
	When intended for such use, the supply disconnecting device shall meet the colour requirements of 10.7.4		P
10.8	Devices for emergency switching off		P
	Devices for emergency switching off will be located separate from operator control stations	Pass muster	P
10.8.1	Location		P
10.8.2	Types	A push-button operated switch	P
	The devices shall be of the self-latching type and shall have positive opening operation	Pass muster	P
	The push-button operated switch may be in a break-glass enclosure		P
10.8.3	Restoration of normal function after emergency switching off		P
	It shall not be possible to restore an emergency switching off circuit until the emergency switching off device has been manually reset		P
10.8.4	Actuators		P
	Actuators of emergency switching off devices shall be coloured RED		P
10.8.5	Local operation of the supply disconnecting device to effect emergency switching off		P
	Where the supply disconnecting device is to be locally operated for emergency switching off, it shall be readily accessible and should meet the colour requirements of 10-8-4		P
10.9	Displays		P
	Displays shall be selected and installed in such a manner as to be visible from the normal position of the operator	Pass muster	P
	Where displays are intended to be warning devices, it is recommended that they be of the flashing or rotary type and be provided with an audible warning device		P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict

11	**Electronic equipment**		**P**
11.1	General		P
	This clause applies to all types of electronic devices including programmable electronic equipment, subassemblies, printed circuit boards, devices and components		P
11.2	Basic requirements		P
11.2.1	Inputs and outputs		P
	An indication of the status of all digital inputs and outputs should be provided	This function has been provided	P
11.2.2	Equipotential bonding		P
	All input/output racks, processor racks, and power supplies shall be electrically bonded together in accordance with the supplier's specifications and connected to the protective	According to the supplier's specifications	P
11.3	Programmable equipment		P
11.3.1	Programmable controllers		P
	Programmable controllers shall conform to relevant IEC standards	The controller used on these machines is in conformity with the requirements of EMC standards	P
11.3.2	Memory retention and protection		P
	Means shall be provided to prevent memory alteration by unauthorized persons and the requirements detailed in 9.4.3.2 shall apply	The modification of the controller is only possible by authorized persons and the requirement in 9.4.3.2 has been applied	P
11.3.3	Software verification		P
	Equipment using reprogrammable logic shall have means for verifying that the software is in accordance with the relevant program documentation	This function has been provided	P
11.3.4	Use in safety-related functions		P
	Programmable electronic equipment shall not be used for category 0 emergency stop functions	This requirement has been complied with	P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict

12	**Controlgear: location, mounting, and enclosures**		
12.1	General requirements		P
	All controlgear shall be located and mounted so as to facilitate:		P
	- its accessibility and maintenance		P
	- its protection against the external influences or conditions under which it is intended to operate		P
	- operation and maintenance of the machine and its associated equipment		P
12.2	Location and mounting		P
12.2.1	Accessibility and maintenance		P
	All items of controlgear shall be placed and oriented so that they can be identified without moving them or the wiring	All of them can be identified without moving or the wiring	P
	All controlgear shall be mounted so as to facilitate its operation and maintenance from the front		P
	Where access is required for regular maintenance or adjustment, the relevant devices shall be arranged between 0,3m and 2,0 m above the servicing level		P
	No devices except devices for operating, indicating, measuring, and cooling shall be mounted on doors and on normally removable access covers of enclosures		P
	Where control devices are connected through plug-in arrangements, their association shall be made clear by type, marking or reference designation, singly or in combination		P
	Plug-in devices that are handled during normal operation shall be provided with non-interchangeable features where the lack of such a facility can result in mulfunctioning	Pass muster	P
	Plug/socket combinations that are handled during normal operation shall be located and mounted so as to provided unobstructed access		P
12.2.2	Physical separation or grouping		P
	Non-electrical parts and devices not directly associated with the electrical equipment shall not be located within enclosures containing control gear	No this kind of parts or devices are located within enclosures containing control gear	P
	Devices such as solenoid valves should be separated from the other electrical equipment	No such solenoid valves	N
	Control devices mounted in the same location and connected to the supply voltage, or to both supply and control voltages, shall be grouped separately from those connected only to the control voltages	Appropriate separation has been taken	P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Terminals shall be separated into groups for: - power circuits; - associated control circuits; - other control circuits, fed from external sources	They have been separated appropriately	P
	The clearances and creep distances specified for the devices shall be maintained	Appropriately clearances and creep distances have been provided	P
12.2.3	Heating effects	No heating effects	N
	Heating generating components shall be so located that the temperature of each component in the vicinity remains within the permitted limit		N
12.3	Degrees of protection		P
	The minimum degree of protection is IP40 for enclosures of switching devices of sewing units and systems		P
12.4	Enclosures, doors and openings		P
	Enclosure shall be constructed using materials capable of withstanding the mechanical, electrical and thermal stresses	The material (metal plate with painting) used for enclosure can withstand the mechanical, electrical and thermal stresses	P
	Fasteners used to secure doors and covers should be of the captive type	Captive type	P
	Windows provided for viewing internally mounted indicating devices shall be of a material suitable to withstand mechanical stress and chemical attach	No this kind of window	N
	It is recommended that enclosures doors shall have: - not wider than 0.9m - vertical hinges - lift-off type - angle of opening at least 95°	These requirement have been taken	P
	If enclosures which readily allow a peraon fully to enter, the relevant requirements specified in this clause shall be comply	No this kind of situation	N
	The joints or gaskets of doors, lids, covers and enclosures shall withstand the chemical effects of the aggressive liquids, vapours used on the machine	They can withstand the chemical effects of the aggressive liquids, vapours, or gases used on the machine	P
	The means used to maintain the degree of protection of an enclosure on doors, lids and covers that require opening or removal for operation or maintenance shall be secured	They can be secured firmly	P
	The degree of protection for all openings in the enclosures shall be secured	The degree of protection can be secured	P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Openings for cable shall be easily re-opened on site	The can be re-opened easily	P
	There shall be no opening between enclosures containing electrical equipment and compartments containing coolant, lubricating or hydraulic fluids, or those into which oil, other liquids, or dust can penetrate	No this kind of opening has been found	N
	The requirement mentioned above does not apply to electrical devices specially designed to operate in oil nor to electrical equipoment in which coolants are used	No applicable	N
	Where there are holes in an enclosure for mounting purpose, the degree of protection for the enclosure shall be secured	Appropriate protection degree can be secured	P
	Equipment that, can attain a suiface temperature sufficient to cause a risk of fire or harmful effect to an enclosure material, the relevant requirements shall be complied	No this kind of gangway has been found	N
12.5	Access to controlgear		P
	The minimum dimensions of gangways in front of and between controlgear shall be in accordance with 481.2.4 of IEC 60364-4-481		P
13	**Conductors and cables**		P
13.1	General requirements		P
	Conductors and cables shall be selected so as to be suitable for the operating conditions current, protection against electric shock, grouping and external influences	All of conductors and cables used on these machines are suitable for the operating conditions and external influences	P
13.2	Conductors		P
	Conductors shall be of copper	Copper	P
	Conductors of any other material shall have a nominal cross-sectional area such that, carrying the same current, the max, temerparure shall not exceed the value given in table 4	Only copper conductors are used	N
	If aluminium is used, the cross-sectional area shall be at least 16mm^2	Only copper conductors are used	N
	All conductors that are subject to frequent movement shall have flexible stranding of class 5 or class 6 (see table c.4)		P
13.3	Insulation		P
	The dielectric strength of the insulation shall be adequate for the test voltage required with a minimum of 2000V a.c. for 5min duration for cables operating at voltages higher than 50V a.c. or 120V d.c	This test has been carried out for the cables, and there is no breakdown is occurred	P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The mechanical strength and thickness of the insulation shall be such that the insulation cannot be damaged in operation or during laying, especially for cables pulled into ducts	Appropriate insulation with sufficient mechanical strength and thickness is provided	P
13.4	Current-carrying capacity in normal service		P
	The cross-sectional area of a conductor shall be such that, under those conditions, the conductor temperature does not exceed the value given in table 4, unless otherwise specified by the cable manufacturer	No exceed temperature	P
	The current-carrying capacities for PVC insulated wiring between enclosure and individual items of equipment under steady0state conditions are given in Table 5		P
13.5	Conductor and cable voltage drop		P
	The voltage drop from the point of supply to the load shall not exceed 5% of the nominal voltage under normal operating conditions	Not exceed 5%	P
13.6	Minimum cross-sectional area		P
	To ensure adequate mechanical strength, the cross-sectional area of conductors should not be less than as shown in Table 6		P
13.7	Flexible cables		P
13.7.1	General		P
	Flexible cables shall have class 5 or class 6 conductors		P
	Cables that are subjected to server duties shall be of adequate construction		P
13.7.2	Mechanical rating		P
	The tensile stress for copper conductors shall not exceed 15N/mm^2 of the copper cross-sectional area		P
	If the demands of the application exceed the tensile stress, it of 15N/mm^2 , cables with special construction fertures should be used and the allowed max, tensile stress strength should be agree with the cable manufacturer		P
13.7.3	Current-carrying capacity of cables wound on drums		P
	Cables to be wound on drums shall be selected with conductors having a cross-sectional area such that, when fully wound on the drum and carrying the normal service load, the maximum allowable conductor temperature is not exceeded	Not exceeded	P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
13.8	Collector wires, collector bars and slip-ring assemblies		P
13.8.1	Protection against direct contact		N
	Collector wires, collector bars and slip-ring assemblies shall be installed or enclosed by the applicantion of one of the following protective measures: - by partial insulation of live parts - by enclosures or barriers of at least IP2X		N
	Min. Protector degree of horizontal top surface of barriers or enclosures that are readily accessible:IP4X		N
	If the required degree of protection is not achieved, protection by placing live parts out of reach in combination with emergency swetching off according to 9.2.5.4.3 shall be applied		N
	Collector wires and collector bars shall be so placed and/or protected as to : - prevent contact - prevent damage from a swinging load		N
13.8.2	Protective conductor circuit		P
	The protective conductor(PE) and neutral conductor (N) shall each use a separate collector wise, collector bar or slip-ring	Pass muster	P
	The continuity of the protective conductor circuit using sliding contacts shall be ensured by taking appropriate measures		P
13.8.3	Protective conductor current collectors		P
	Protective conductor current collectors shall have a shape or construction so that they are not interchangeable with the other current collectors		P
	Such current collectors shall be of the sliding contact type		P
13.8.4	Removable current collectors with a disconnector function	No applicable	N
	Removable current collectors having a disconnector function shall be so designed that the protective conductor circuit is interrpted only after the live conductors have been disconnected ,and the continuity of the protective conductor circuit is re-estabilished before any live conductor is reconnected		N
13.8.5	Clearance in air	No applicable	N
	Shall be sutable for operation in pollution degree 3 conditions		N

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
13.8.6	Creepage distances		P
	Shall be suitable for operation in pollution degree 3 conditions		P
13.8.7	Conductor system sectioning		P
	If collector wires or collector bars can be divided into isolated sections, suitable design measures shall be employed to prevent the energization of adjacent sections by the current collectors themselves		P
13.8.8	Construction and installation of collector wire, collector bar systems and slip-ring assemblies	No applicable	N
	Collector wire, collector bar systems and slip-ring assemblies used for power circuits shall be grouped separately from those used for control circuits		N
	Removable covers for collector wire and collector bar sustems laid underground or underfloor shall be so designed that they cannot be opened by one person without the aid of a tool		N
	Undergroup and underfloor collector bar ducts shll have drainage facilities		N
14	**Wiring practices**		P
14.1	Connections and routing		P
14.1.2	Conductor and cable runs		P
	Conductor and cables shall be run from terminal to terminal without splices or joints	All of them are run from terminal to terminal without splices or joints	P
	Where it is necessary to connect and disconnect cables and cable assemblies, a sufficient extra length shall be provided for that purpose		P
	The terminations of cables shall be adequately supported to prevent mechanical stresses at the terminations of the conductors	Adequate support measure has been taken	P
14.1.3	Conductors of different circuits	No conductors of different circuits	N
	Conductors of different circuits may be laid side by side, may occupy the same duct, or may be in the same multiconductor cable provided that the arrangement does not impair the proper functioning of the respective circuits		N
14.2	Identification of conductors		P
14.2.1	General requirements		P
	Conductors shall be identifiable at each termination in accordance with the technical documentation	Make reference to clause 18	P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Use of color-coding for identification of conductors	Color-coding for identification is used	P
	Color GREEN or YELLOW should hot be used	No GREEN or YELLOW conductor is used	P
14.2.2	Identification of the protective conductor		P
	The protective conductor shall be readily distinguishable by shape, location, marking ,or color	By marking and color	P
	For insulated conductors, the bicolour combination GREEN-AND-YELLOW shall be such that on any 15mm length one of the colours covers at least 30% and not more than 70% of the surface of the conductor, the other covering the remainder of the surface		P
	Use of graphical symbol: ⏚		P
14.2.3	Identification of the neutral conductor	No neutral conductor is used	N
	Where a circuit includes a neutral conductor identified by colour, the colour shall be LIGHT BLUE		N
14.2.4	Identification of other conductors		N
	Identification of other conductors shall be by colour, number, alphanumeric, or a combination of colour and numbers or alphanumeric		N
	Conductors used for functional earthing shall be identified by the colour GREY		N
	Common conductors, for example for eliminating static charges, shall be identified by colour GREY		N
14.3	Wiring inside enclosures		P
	Panel conductors shall be supported where necessary to keep them in place	Appropriate supports are provided	P
	Non-metallic ducts shall be permitted only when they are made with a flame-retardant insulating material	Some non-metallic ducts are used with a flame-retardant insulating material	P
	Connections to devices mounted on doors or to other movable parts shall be made using flexible conductors according to 13.2		P
	The conductors shall be anchored to the fixed part and to the movable part independently of the electrical connection	Adequate anchored measures have been taken	P
	Conductors and cables that do not run in ducts shall be adequately supported	All of them have been supported adequately	P
	Terminal blocks or plug/socket combinations shall be used for control wiring that extends beyond the enclosure	This application has been taken	P
14.4	Wiring outside enclosures	Not wiring outside enclosures	N

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
14.4.1	General requirements		N
	The means of introduction of cables or ducts with their individual glands, bushings, etc., into an enclosure shall ensure that the degree of protection is nor reduced		N
14.4.2	External ducts		N
	Fitting used with ducts or multiconductor cable shall be suitable for the physical environment		N
	Flexible conduit or flexible multiconductor cable shall be used for connections involving small or infrequent movements		N
	The weight of the pendant stations shall be supported by means other that the flexible conduit or the flexible multiconductor cable		N
	Flexible conduit or flexible multiconductor cable shall be sued for connections involving small or infrequent movements		N
14.4.3	Connection to moving elements of the machine		N
	Connections to frequently moving parts shall be made using conductors in accordance with 13.2		N
	Cables subject to movement shall be supported in such a way that there is no mechanical strain on the connection points nor any sharp flexing		N
	The cable sheath shall be resistant to the normal wear that can be expected from movement and to the effects of atmospheric contaminants		N
	Where cables subject to movement are close to moving parts ,precautions shall be taken to maintain a space of at least 25mm between the moving parts and the cables		N
	The cable handling system shall be so designed that lateral cable angles do not exceed 5° , avoiding torsion in the cable :test		N
	Devices serving to guide and carry a flexible cable shall be so designed that the inner bending radius at all points where the cable is bent is not less than the values given in Table 8		N
14.4.4	Interconnection of devices on the machine		N
	Such terminals shall be conveniently placed, adequately protected, and shown on the relevant diagrams		N

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
14.4.5	Plug/socket combinations		N
	Where equipment is removable, connections to is through a polarized plug/socket combination are permitted		N
	Plug/socket combinations shall be of such a type and so installed to prevent unintentional contact with live parts at any time even during insertion or removal of the connectors		N
	Plug/socket combinations that are rated at more than 16A or that remain connected during normal service shall be of a retaining type to prevent unintended disconnection		N
	Where more than one plug/socket combination is used in the same electrical equipment, they shall be clearly identifiable		N
14.4.6	Dismantling for shipment		N
	Plug/socket combinations shall be protected from the physical environment during transportation and storage		N
14.4.7	Additionsal conductors		N
	Spare conductors shall be connected to spare terminals or isolated to prevent contact with live parts		N
14.5	Ducts, connection boxes and other boxes		P
14.5.1	General requirements		P
	Ducts shall provide a minimum degree of protection of IP33		P
	Appropriate protection for conductors insulation	Suitable protection is taken	P
	Drain holes of 6mm diameter are permitted		P
	Ducts and cables trays shall be rigidly supported and positioned at a sufficient distance from moving parts	Suitable support and sufficient distance have been taken	P
	In areas where human passage is required, the ducts and cable trays shall be mounted at least 2m above the working surface	No this kind of area	N
	Ducts shall be provided only for mechanical protection	Adequate mechanical protection is provided	P
	Cable trays that are partially covered should not be condidered to be ducts or cable trunking system, and the cables used shall be suitable for installation on cable trays		P
14.5.2	Percentage fill of ducts		P
	The dimensions and arrangement of the ducts be such as to facilitate the insertion of the conductors and cables	This requirement has been complied with	P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
14.5.3	Rigid metal conduit and fittings	No rigid metal conduit is used	N
	Rigid metal conduit and firings shall be of galvanized steel or of a corrosion-resistant material suitable for the conditions		N
	Conduits shall be securely held in place and supported at each end		N
	Fittings shall be compatible with the conduit and appropriate for the application		N
	Conduit bends shall be made in such a manner that the conduit shall not be damaged and the internal diameter of the conduit shall not be effectively reduced		N
14.5.4	Flexible metal conduit and fittings	No rigid metal conduit is used	N
	A flexible metal conduit shall consist of a flexible metal tubing or woven wire armour		N
	Fittings shall be compatible with the conduit and appropriate for the application		N
14.5.5	Flexible non-metallic conduit and fittings	No flexible non-metal conduit is used	N
	A flexible non-metallic conduit shall be resistant to kinking and shall have physical characteristics similar to those of the sheath of multiconductor cables		N
14.5.6	Cable trunking systems	No cable trunkling system is used	N
	Cable trunking systems external to enclosures shall be rigidly supported and cleat of all moving or contaminating portions of the machine		N
	Covers shall be shaped to overlap the sides; gaskets shall be permitted		N
	On horizontal cable trunking systems, the cover shall not be on the bottom		N
	Where the cable trunking system is furnished in sections, the joints between sections shall fit tightly but need not be gasketed		N
	The only openings permitted shall be those required for wiring or for drainage		N
14.5.7	Machine compartments and cable trunking systems	No this kind of situation	N
	Are isolated from coolant or oil reservoirs and are entirely enclosed		N
	Conductors run in enclosed compartment and cable trunking systems shall be so secured and arranged that they are not subject to damage		N
14.5.8	Connection boxes and other boxes		P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Connection boxes and other boxes used for wiring purposes shall be readily accessible for maintenance	They are readily accessible for maintenance	P
	shall provide protection against the ingress of solid bodies and liquids	Adequate protection is provided	P
	Those boxes shall not have opened but unused knockouts nor any other openings and shall be so constructed as to exclude materials such as dust, flyings, oil, and coolant	These requirements have been complied with	P
14.5.9	Motor connection boxes		P
	Motor connection boxes shall enclose only connections to the motor and motor-mounted devices	They enclose only connections to the motor and motor-mounted devices	P
15	**Electric motors and associated equipment**		N
15.1	General requirements		N
	Electric motors should conform to the requirements of IEC60034-1		N
	Motor control equipment shall be located and mounted according to clause 12		N
15.2	Motor dimensions		N
	It is recommended that motor enclosures be chosen from those included in IEC60034-5		N
	The degree of protection shall be at least IP23 for all motors		N
	The minimum degree of protection of the sewing machine drive shall be IP40		N
15.3	Motor dimensions		N
	As far as is practicable, the dimensions of motors shall conform to those given in IEC 6072-1 and IEC60072-2		N
15.4	Motor mounting and compartment		N
	Each motor and its associated couplings, belts and pulleys, or chains, shall be so mounted that they are adequately protected and are easily for inspection		N
	Shall be such that all motor hold-down means can be removed and all terminal boxes are accessible		N
	The proper cooling shall be ensured and the temperature rise remains within the limits of the insulation class		N
	Motor compartment should be clean and dry, and shall be ventilated directly to the exterior of the machine		N
	The vents shall be such that ingress of swarf, dust, or water spray is at an acceptable level		N

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	There shall be no opening between the motor compartment and any other compartement that does not meet the motor compartment requirements		N
	If a conduit or pipe is run into the motor compartment from another compartment not meet the motor compartment requirements, any clearance around the conduit or pipe shall be sealed		N
15.5	Criteria for motor selection		N
	Shall be selected according to the anticipated service and physical environment conditions		N
15.6	Protective devices for mechanical brakes		N
	Operation of the overload and overcurrent protective devices for mechanical brake actuators shall initiate the simultaneous de-energization of the associated machine actuators		N
16	**Accessories and lighting**		**N**
16.1	Accessories		N
	Socket-outlets for accessory equipment shall comply:	Not such socket-outlets	N
	Should conform to IEC 60309-1 (if this is not possible, they should be clearly marked with the voltage and current ratings)		N
	The continuity of the protective bonding circuit to the socket-outlet shall be ensured		N
	All unearthed conductors: over current or overload protection according to 7.2 and 7.3 separately from the protection of other circuits		N
	If the power supply to the socket outlet is not disconnected by the supply disconnecting device, the clause 5.3.5 shall apply		N
16.2	Local lighting of the machine and equipment	No such local lighting	N
	Connections to the protective bonding circuit shall be in accordance with 8.2.2		N
	The ON-OFF switch shall not be incorporated in the lampholder or in the flexible connecting cords		N
	Stroboscopic effects from lights shall be avoided by the use of appropriate luminaires		N
	Where fixed lighting is provided in an enclosure, electromagnetic compatibility should be taken into account using the principles outlined in 4.4.2		N
16.2.2	Supply		N

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	It is recommended that the nominal voltage of the local lighting circuit should not exceed 50V between conductors		N
16.2.3	Protection		N
	Operators during circuits shall be protected in accordance with 7.2.6		N
16.2.4	Fittings		N
	Adjustable lighting fittings shall be suitable for the physical environment		N
	Reflectors shall be supported by a bracket and not by the lampholder		N

17	**Marking, warning signs and reference designations**		**P**
17.1	General		P
	The electrical equipment shall be marked with the supplier's name ,trade mark, or other identifying symbol and, when required, with a certification mark	These information have been marked	P
	Warning signs, nameplates, markings, and identification plates shall be of sufficient durability to withstand the physical environment involved	They can withstand the physical environment involved	P
17.2	Warning signs		P
	Enclosures shall be marked with the warning sign: 	This warning sign has been used	P
	The warning sign shall be plainly visible on the enclosure door or cover	It is plainly visible on the enclosure door	P
17.3	Functional identification		P
	Control devices, visual indicators and displays, used in man-manchine interface shall be clearly and durably marked with regard to their functions either on or adjacent to the item	Appropriate markings have been provided for these devices	P
	Preference should be given to the use of standard symbols give in IEC 60417 and ISO 7000	These relevant requirements appropriate for this machine have been used	P
17.4	Marking of control equipment		P
	Control equipment shall be legibly and durably marked in a way that is plainly visible after the equipment is installed	They have been marked legibly and durably	P
	The full-load current shown on the nameplate shall be not less than the combined full-load currents for all motors and other equipment that can be in operation at the same time under normal conditions of use	A nameplate is used	P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Where only a single motor controller is used, that information may instead be provided on the machine nameplate where it is plainly visible		P
17.5	Reference designations		P
	All enclosures, assemblies, control devices, and components shall be plainly identified with the same reference designation as shown in the technical documentation that shall be in accordance with IEC61346-1	These information have been provided within the instruction manual	P
	Where size or location preclude the use of an individual reference designation, group reference designation shall be used		P
18.	**Technical documentation**		P
18.1	General		P
	The information necessary for installation, operation, and maintenance of the electrical equipment of a machine shall be supplied in the form of drawing ,diagrams, charts, tables ,and instructions	All the information have been provided by many forms	P
	The information shall be in an agreed language	English	P
	The supplier shall be ensure that the technical documentation in this clause is provided with each machine	The instruction manual is equipped with each machine	P
18.2	Information to be provided		P
	The information provided with the electrical equipment shall include the requirements specified in this clause		P
18.3	Requirements applicable to all documentation		P
	Relevant requirements according to 18.4 to 18.10 shall be complied		P
18.4	Basic information		P
	Min. requirements for the technical documentation shall be contained		P
18.5	Installation diagram		P
	The recommended position, type, and cross-sectional areas of the supply cables to be installed on site shall be clearly indicated		P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The size, and purpose of ducts, cable trays, or cable supports between the machine and the associated equipment that are to be provided by the user shall be detailed		P
18.6	Block(system) diagrams and function diagrams		P
	Use and requirements for system(block) diagram	Installation diagrams are provided	P
18.7	Circuit diagrams		P
	Use and requirements for circuit diagrams	Circuit diagrams are provided	P
18.8	Operating manual		P
	The technical documentation shall contain an operating manual detailing proper procedures for set-up and use of the equipment		P
	Where the operation of the equipment can be programmed, detailed information on methods of programming, equipment required, program verification, and additional safety procedures shall be provided		P
18.9	Maintenance manual		P
	The technical documentation shall contain a maintenance manual detailing proper procedures for adjustment, servicing and prevnetive inspection, and repair	Maintenance manual is proveded	P
18.10	Parts list		P
	Use and requirements for parts list	Parts list is provided	P

19.	**Testing and verification**		P
19.1	General		P
	When these tests are performed, it is recommended that they follow the sequence listed	All tests have been carried out according to the following sequence	P
	When the electrical equipment is modified, the requirements stated in 19.7 shall apply		P
19.2	Continuity of the protective bonding circuit		P
	Test conditions: a current of at least 10A at 50Hz OR 60Hz		P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The measured voltage shall not exceed the values given in table 9		P
19.3	Insulation resistance tests		P
	Test conditions : 500 V d.c		P
	The measured values shall not less than 1MΩ	See the test report in detail	P
19.4	Voltage tests		P
	Test conditions: - at least 1 second - test voltage is twice the raged supply voltage of the equipment or 1000V, whichever is greater - frequency of 50/60Hz - supplied from a transformer with a min.rating of 500V/A		P
	Shall not breakdown	See the test report in detail	P
19.5	Protection against residual voltages		N
	Tests are performed to ensure compliance with 6.2.4		N
19.6	Functional tests		P
	The functions of electrical equipment shall b tested ,particularly those related to safety and safeguarding	All functions equipped with this machine have been tested	
19.7	Retesting		N
	Where a portion of the machine and its associated equipment is changed or modified, that portion shall be reverified and retested, as is appropriate		N

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict

Annex			P
Annex A	**examples of machines covered by this part of IEC 60204**		P
Annex B	**inquiry form for the electrical equipment of machines**		P
Annex C	**current-carrying capacity and overcurrent protection of conductors and cables in the electrical equipment of machines**		P
	The purpose of this annex is to provide additional information on the selection of conductor sizes where the conditions given Table 5 have to be modified		P
Annex D	**explanation of emergency operation functions**		P
	Emergency stop		P
	Emergency start		P
	Emergency switching off		P
	Emergency switching on		P
Annex E	**bibliography**		P

Annex			P
Annex A	**examples of machines covered by this part of IEC 60204**		P
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	Emergency stop		P
	Emergency start		P
	Emergency switching off		P
	Emergency switching on		P
Annex E	**bibliography**		P

US IEC 60335-2-45;US IEC 60335-1			
Clause	Requirement + Test	Result - Remark	Verdict

19.3	**Insulation resistance measurements**		**P**
Insulation resistance R between:		Measurement(MΩ)	Requirement(MΩ)
Between the power circuit conductors and the protective bonding circuit		5	1

19.4	**TABLE: electric strength measurements**		**P**
Test voltage applied between:		Test voltage (V)	Breakdown
Between the conductors of all circuits and the protective bonding circuit		AC 1,000V rms	No

Photograph Document

